Course Code	Course Title	L	T	P	C
MACSE518	**Database Modeling and Design**	**3**	**0**	**2**	**4**
Pre-requisite	Nil	**Syllabus Version**			
		1			

Course Objectives

1. To inculcate the foundational knowledge of database concepts, ER modeling, relational algebra, and optimization techniques for effective database design.
2. To equip with advanced skills in EER modeling, complex relationship modeling, schema mapping, and transactional programming using SQL and PL/SQL.
3. To impart skills in normalization, distributed systems, advanced data modeling, and cutting-edge NoSQL and Big Data technologies.

Course Outcomes

1. Comprehend the fundamental of database systems and its architectures.
2. Model the database systems using ER and EER to design complex database systems.
3. Analyze and apply normalization theory to relational schemas and optimize the design for minimal redundancy.
4. Gain insight into advanced database technologies for handling complex problems.
5. Design NoSQL databases and handle big data to solve real-world problems.

Module:1	**Introduction to Database Systems**	**6 hours**

Data, Information, Metadata, Data Management - Purpose of Database– Data Models and concepts- Database System Components - Three Schema Architecture– Centralized and Client/Server Architectures–Conceptual Design using ER Modeling: Entities, Attributes, Relationships, Integrity Constraints, Strong Entities, Weak Entities– ER Modeling Process.

Module:2	**Database Modeling and Database Languages**	**10 hours**

Enhanced Entity-Relationship (EER) Modeling: Subclasses, Superclasses, and Inheritance– Specialization and Generalization– Constraints– Modeling of UNION Types Using Categories- Ternary and Higher Order Relationships, Recursive Relationships– Mapping ER and EER Models to Relational Schema– Introduction to Relational Algebra: Basic Operators: Selection, Projection, Join, Union, Intersection– Views and Materialized Views– Query Optimization– Indexing. Database Languages: Data Definition Language (DDL)– Data Manipulation Language (DML)– Data Control Language (DCL)– Display Queries (DQL)– Joins, Subqueries– Transaction Control Language (TCL)– COMMIT, ROLLBACK, SAVEPOINT– PL/SQL– Cursors, Triggers.

Module:3	**Relational Database Design and Transaction Management**	**10 hours**

Normalization – Functional Dependencies: Concept, Equivalence Set– Minimal Cover, Attribute Closure– Deriving Candidate Keys and Purpose of Keys– Normal Forms: 1NF, 2NF, 3NF, BCNF– Comprehensive Normalization Examples– Dependency Preserving Decomposition– Lossless Join Decomposition– Higher Normal Forms: Multi-valued Dependencies, 4NF, Join Dependencies, 5NF, Domain-Key Normal Form (DKNF) –Transactions – Concepts of Transactions– Serializability– Concurrency Control – Database Recovery.		
Module:4	**Advanced Database Systems**	**10 hours**
Parallel Databases: Architecture, Interoperation Parallelism and Intraoperation parallelism – Data Partitioning Strategies– Query Optimization for Parallel Execution– Distributed Database Concepts– Features– Fragmentation, Allocation, and Replication– Concurrency Control and Recovery in Distributed Databases– Query Processing and Optimization in Distributed Databases– Spatial Database– Temporal Database– Multimedia Database Concepts– Graph Databases - Knowledge Graphs.		
Module:5	**NoSQL Databases and Big Data Systems**	**7 hours**
Semi-Structured Data Models: JSON, XML, RDF– NoSQL Databases: MongoDB, Key–Value Stores– Big Data Systems: HDFS Overview, Apache Spark– Cloud Data Storage.		
Module:6	**Contemporary Issues**	**2 hours**
	Total Lecture Hours:	**45 hours**
Text Book(s)		
R. Elmasri and S. Navathe, "**Fundamentals of Database Systems**", Addison-Wesley, 7th Edition, **2021** D Abraham Silberschatz, Henry F. Korth, S. Sudarshan , "**Database System Concepts**", McGraw Hill, 7th Edition, **2021**		
Reference Books		
Edward Sciore, "**Database Design and Implementation: Second Edition (Data-Centric Systems and Applications)**", Springer, 2nd Edition, **2020** Sadalage & Fowler, "**NoSQL Distilled: A Brief Guide to the Emerging World of Polyglot Persistence**", Addison-Wesley, 2nd Edition, **2021** Guy Harrison, "**Next Generation Databases: NoSQL, NewSQL, and Big Data**", Apress, 2nd Edition, **2020**		
Indicative Experiments		
1. Design an EER diagram for a Research Collaboration Portal. • Entities: Professors, Students, Projects, Publications, FundingAgencies. • Generalization: Person (Professor, Student).		2 hours

• Specialization: Professors can be Principal Investigator (PI) or Co-Investigator (Co-PI). • Recursive Relationship: Professors mentoring junior faculty or students. • Ternary Relationship: A project is supervised by a PI, funded by a FundingAgency, and has at least one student. • Constraints: One project cannot exceed ₹10,00,000 in funding per year. A student cannot be on more than 2 active projects. Cardinality, Participation Tools: Lucidchart, Draw.io, MySQL Workbench.	
2. Relational Schema Mapping: Transform the EER diagram from lab 1 into its corresponding relational schema representation. List all resulting tables with primary keys, foreign keys, and attributes. • Create and populate database tables using appropriate DDL and DML commands. • Specify how the following are handled: Superclass-subclass mapping strategy (e.g., one table per subclass). Recursive relationship. Ternary relationship. Constraints (primary key, foreign key, check, default, not null, unique). Platform: MySQL / PostgreSQL / Oracle.	2 hours
3. Execute advanced SQL queries on the relational schema derived from the EER diagram. This includes retrieving, filtering, and analyzing data using: • Aggregate functions (SUM, AVG, MAX, MIN, COUNT) • GROUP BY, HAVING clauses • Nested subqueries (scalar, correlated, multi-row) • Window functions (ROW_NUMBER, RANK) Platform: MySQL / PostgreSQL / Oracle.	2 hours
4. Develop and execute advanced SQL queries on the relational schema derived from the EER diagram, focusing on advanced data retrieval, filtering, and analysis. This includes the use of: • Standard Joins: Inner Join, Outer Join (Left, Right, Full), Self-Join • Special Joins: Cross Join, Natural Join • Other Joins: Equi Join, Non-Equi Join, Semi Join, Anti Join (using EXISTS / NOT EXISTS) Platform: MySQL / PostgreSQL / Oracle.	2 hours

5. DCL and TCL operations on the relational schema derived from the Research Collaboration Portal. This includes managing user permissions and ensuring consistency of transactional operations using: • DCL Commands: GRANT, REVOKE to control access privileges on tables such as Professors, Projects, and Publications. • TCL Commands: COMMIT, ROLLBACK, and SAVEPOINT to manage and recover database states during multi-step operations like funding updates or project allocations. Platform: MySQL / PostgreSQL / Oracle.	2 hours
6. Using the relational schema of the Research Collaboration, perform the following tasks: • Create indexes based on query requirements. • Define sequences for automatically generating unique IDs (Oracle uses sequences, MySQL uses AUTO_INCREMENT, and PostgreSQL supports both (SERIAL / SEQUENCE)). • Create views to simplify common queries. Platform: MySQL / PostgreSQL / Oracle.	2 hours
7. PL/SQL – Procedures and Functions: Implement and execute stored procedures and functions. Using the Research Collaboration Portal relational schema, implement and execute stored procedures and functions to automate and encapsulate common operations. Tasks: • Create a stored procedure to add a new project. The procedure should accept project details (title, PI ID, funding agency ID, funding amount, start and end dates) and validate that the funding amount does not exceed ₹10,00,000. If the validation fails, rollback the transaction and raise an appropriate error. • Create a function to count the number of active projects for a given student ID. The function should return the count of projects where the current date falls between the project start and end dates. • Create a stored procedure to assign a student to a project only if the student is currently involved in fewer than 2 active projects. If the student is eligible, insert the assignment; otherwise, raise an error. • Create a function that returns the total funding amount received by a specific professor (as PI) across all projects. Platform: MySQL / PostgreSQL / Oracle.	2 hours
8.	2 hours

PL/SQL – Cursors and Triggers: Using the Research Collaboration Portal schema, implement and execute PL/SQL cursors and triggers to handle complex data retrieval and enforce data integrity. Tasks: • Write an explicit cursor to fetch and display all projects along with their Principal Investigator's name and total funding amount. Include cursor open, fetch, and close operations with appropriate exception handling. • Use an implicit cursor to count the number of publications authored by each professor and display the results. • Create a BEFORE INSERT trigger on the Projects table to validate that the funding amount does not exceed ₹10,00,000. If the validation fails, raise an application error to prevent insertion. • Create an AFTER INSERT trigger on the Project_Student table to check that a student is not assigned to more than 2 active projects. If violated, raise an error and roll back the insert. • Create a BEFORE UPDATE trigger on the Professors table to prevent changing the designation of a professor from "Principal Investigator" to any other value if they currently supervise active projects. Platform: MySQL / PostgreSQL / Oracle.	
9. Design a MongoDB-based Student Management System where: • Each student has a unique ID, name, age, and department. • Each student is enrolled in multiple courses. • Each course includes course_code, course_name, credits Platform: MongoDB	2 hours
10. Spatial Databases using GIS Coordinates: Design a spatial database for managing university campuses, department buildings, and student hostels. Tasks: • Store GPS locations of all buildings • Find hostels within 1 km of academic buildings • Identify the nearest cafeteria to each department • Visualize results using GeoJSON exports Platform: PostgreSQL/PostGIS or MongoDB with GeoJSON	2 hours
11. Build a basic application that interacts with cloud-based databases or Big Data frameworks for scalable data storage, retrieval, and analysis. Tasks: • Connect and interact with a cloud database (e.g., Firebase, RDS, BigQuery) • Perform CRUD operations over the cloud	2 hours

• Integrate data ingestion and processing with Big Data tools (e.g., Spark or Hadoop) • Analyze large datasets in a distributed environment • Understand deployment, scalability, and cloud storage optimization Tools: Firebase Realtime DB / Firestore or AWS RDS / MongoDB Atlas or Google BigQuery / Azure Cosmos DB.	
Total Laboratory Hours:	**30 hours**
Mode of Evaluation : Continuous Assessment Test, Digital Assignment, Quiz, Final Assessment Test, Lab Continuous Assessment, Lab Final Assessment, Presentaion	
Recommended by Board of Studies :	23-05-2025
Approved by Academic Council : No. 78	12-06-2025